

# Labwork 2 - Numba Introduction

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## 1 Check Device with `numba.cuda.detect()`

I have an AMD GPU and Numba-HIP installed, it's supposed to be compatible with `numba.cuda` bindings but it is broken as of now, so I can not print out the information about my GPU. It is a Radeon RX6800.

GPU acceleration in Numba works, I have verified it.

Cuda bindings also somewhat working with HIP by using `numba.hip.pose_as_cuda()`

Results of using `numba.cuda.detect()` is as follows:

```
uv run lab2/lab2.py
```

```
Found 1 CUDA devices
```

```
Traceback (most recent call last):
```

```
  File "/home/tb24/projects/advancedhpc2026/lab2/lab2.py", line 3, in <module>
    device = hip.detect()
            ~~~~~
```

```
  File "/home/tb24/projects/advancedhpc2026/.venv/lib/python3.11/site-
packages/numba/hip/api.py", line 525, in detect
    cc = dev.compute_capability
        ~~~~~
```

```
  File "/home/tb24/projects/advancedhpc2026/.venv/lib/python3.11/site-
packages/numba/hip/hipdrv/devices.py", line 128, in __getattr__
    return getattr(self._device, item)
           ~~~~~
```

```
  File "/home/tb24/projects/advancedhpc2026/.venv/lib/python3.11/site-
packages/numba/hip/hipdrv/driver.py", line 670, in __getattr__
    code = getattr(
           ~~~~~
```

```
  File "/home/tb24/.local/share/uv/python/cpython-3.11.13-linux-x86_64-
gnu/lib/python3.11/enum.py", line 786, in __getattr__
    raise AttributeError(name) from None
AttributeError: CU_DEVICE_ATTRIBUTE_compute_capability
```